

Seung-Hyeon Lim

Undergraduate Researcher | Compiler Engineer | Android Native (Mobile Application) Developer

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SUMMARY

- Undergraduate researcher specializing in LLVM-based compiler optimization for FPGA High-Level Synthesis
- Designed and implemented a custom LLVM pass achieving 39.4% average LUT reduction on CHStone benchmarks, with cryptographic kernels previously unimplementable on target hardware made feasible through IR-level bit-width transformation
- Recognized at KSC 2025 (Encouragement Award). Fluent across the full C-to-RTL compilation pipeline and motivated to deepen expertise in compiler backends, program analysis, and hardware-software co-design

EDUCATION

- Kyung Hee University – B.S. in Computer Science and Engineering**
 - Period : Mar. 2021 – Present (Expected Feb. 2027)
 - GPA : 3.745 / 4.3 (4.04 / 4.5)

RESEARCH EXPERIENCE

- Undergraduate Researcher – Compiler and System Software Lab at Kyung-Hee University**
 - Period : Mar. 2025 – Present
 - Advisor: Prof. Seonyeong Heo

Designing and implementing a custom LLVM compiler pass for FPGA resource optimization through bit-level static analysis. The core contribution is a two-phase algorithm: Enhanced Bit-Usage Analysis via backward dataflow traversal of LLVM IR def-use chains, followed by Data Width Optimization that reconstructs IR with minimal-width operators. Integrated LLVM's Known-Bits API to extend the analysis to non-constant values, enabling application to real-world HLS workloads.

Please find below the key contributions:

- Average **39.4% LUT reduction** and **29.9% latency improvement** on CHStone benchmarks (Xilinx Arty A7-100T, 100 MHz)
- SHA: **74.4% LUT reduction** (259% → 66% utilization) – enabled a previously infeasible design to meet timing constraints.
- Blowfish: **47.6% LUT reduction**, 45.3% latency improvement; pipeline interval reduced from 32 to 9 cycles/iteration.
- Optimization effectiveness correlated with bitwise operation density (SHA: 51.6%, Blowfish: 45.5%); validates hypothesis for bit-intensive workloads.

PUBLICATIONS & AWARDS

- ◆ KSC (Korea Software Congress) 2025 Encouragement Award – Undergraduate/Junior Paper Competition**

- Seung-Hyeon Lim, Seonyeong Heo. "비트 레벨 컴파일러 최적화를 통한 자원 효율적인 FPGA 합성". KSC 2025.

- IEEE Computer Architecture Letters – under submission**

- Seung-Hyeon Lim, Seonyeong Heo. "Resource-Efficient High-Level Synthesis with Bit-Level Compiler Optimization." Achieves 39.4% average LUT reduction on CHStone

- Journal of Systems Architecture, Vol. 168, 2025 – contributing author**

- S. Heo, J. Kim, W. Im, J. Moon, D. Jang, Bit-level compiler optimization for ultra low-power embedded systems, Journal of Systems Architecture 168 (2025) 103546. [doi:10.1016/j.sysarc.2025.103546](https://doi.org/10.1016/j.sysarc.2025.103546).

PROJECTS (SERVICE)

❑ WhiteWand / AEye – Android Application for the Visually Impaired

- ◆ 1st Place, KHU Software Convergence College Club Competition (D.Com, Sep 2022) · 2022 KHU SW Festival AI/Big Data Track
- Participated as Android Developer · github.com/Dcom-KHU/AEye-Android

Developed an Android application enabling visually impaired users to identify pharmaceuticals and beverages via on-device object detection. YOLOv5 model trained on a custom dataset (background injection, rotation, cutout, noise augmentation) converted to TFLite and embedded in the app. Interaction is designed exclusively for accessibility: swipe, volume button, and shake gestures; all feedback delivered via TTS API.

- On-device inference with TFLite-converted YOLOv5 (Kotlin 83.6% / Java 16.4%)
- MVVM architecture: Room local DB, ViewModel data binding
- Custom TextToSpeechManager and ShakeDetector (SensorEventListener) for gesture-driven UX; Undefined-class fallback prevents misclassification

❑ FarmUs – Agricultural Land Rental & Transaction Platform

- UMC 3rd Cohort Developer Matching · Android Developer · github.com/FarmUsProject

Contributed Android client development (Ktor, MVVM, Jetpack) as the team prepared Alpha Version release through UMC's in-program developer matching.

LEADERSHIP & ACTIVITIES

❑ D.Com	Vice President for KHU CS Dept. Academic Club	2022.01 – 2022.12
❑ D.COM-KHU	GitHub Organization Owner	2023.03 – Present
❑ UMC	Uni MakeUs Challenge 3rd Cohort - KHU Android Track Lead	2022.09 – 2023.02
❑ KHU College of SW	Software Membership for Open Source	2026.04 - 2026.11

TECHNICAL SKILLS

❑ Languages	: Kotlin, C/C++, Java, Python, LLVM IR, TypeScript, Dart, OCaml
❑ Compiler / System Software	: LLVM Pass Development, Vitis HLS, FPGA Synthesis (Xilinx)
❑ Service Development	: Android Native Application, Kotlin Multiplatform, Ktor, React, Flutter